

GPS-Based Geofencing and Relay Control System

User Manual

Version 1.0

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1. SYSTEM OVERVIEW

The GPS-Based Geofencing and Relay Control System is designed to restrict vehicle operation based on predefined geographical boundaries. The system continuously monitors vehicle location using a NEO-6M GPS module and processes the coordinates using an ESP32 microcontroller. A virtual geographic boundary (geofence) is created using predefined center coordinates and radius values. When the vehicle remains inside the permitted area, the relay remains activated and the ignition circuit operates normally. If the vehicle moves outside the allowed boundary, the relay automatically disconnects the ignition line.

2. HARDWARE REQUIREMENTS

- ESP32 Development Board
- NEO-6M GPS Module
- 2-Channel Relay Module (Single Channel Used)
- LM2596 Buck Converter
- 12V Vehicle Battery
- PCB Board and Wiring
- 3D Printed Enclosure

3. INITIAL SETUP

- Step 1: Connect the 12V battery to the LM2596 buck converter.
- Step 2: Adjust the output voltage to 5V.
- Step 3: Connect ESP32, GPS module, and relay module to the regulated 5V supply.
- Step 4: Upload the geofencing program through Arduino IDE.
- Step 5: Configure latitude, longitude, and geofence radius as required.

4. HOW TO USE THE SYSTEM

1. Power ON the vehicle.
2. Wait for GPS satellite lock.
3. Verify coordinates through Serial Monitor.
4. Operate the vehicle inside the permitted geofence.

5. If the vehicle exits the geofence, relay control is automatically activated.

5. SYSTEM OPERATION STATES

GPS Searching – Acquiring satellite signals.

Inside Geofence – Relay ON, vehicle enabled.

Outside Geofence – Relay OFF, ignition interrupted.

6. GEOFENCE MONITORING

The system continuously monitors latitude, longitude, distance from the center point, relay status, and geofence condition to determine whether vehicle operation should be permitted.

7. TROUBLESHOOTING

GPS Not Detecting Location – Move vehicle outdoors and wait for satellite lock.

Relay Not Switching – Verify GPIO and power connections.

ESP32 Not Powering – Check LM2596 output voltage.

Incorrect Distance Calculation – Verify coordinate and radius configuration.

8. TECHNICAL SPECIFICATIONS

Input Voltage: 12V DC

Output Voltage: 5V DC

GPS Module: NEO-6M

Controller: ESP32

Relay Module: 5V Dual Channel (Single Channel Used)

Geofence Radius: User Configurable

Enclosure: Custom 3D Printed Housing